Prioritizing Sustainable Protein Sources with Marine Aquaculture

EDS223 HW3

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Setup

```
# Load packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(here)
```

```
here() starts at /Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/Aquacultu
```

```
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(tmap)
```

```
Registered S3 method overwritten by 'stars':
  method                   from
  st_interpolate_aw.stars sf
```

```
library(terra)
```

```
terra 1.9.50

Attaching package: 'terra'

The following object is masked from 'package:tidyr':

    extract
```

```
library(knitr)
```

```

Attaching package: 'knitr'

The following object is masked from 'package:terra':

    spin
```

```
#library(ggpubr)

# Set warnings to display as they occur
options(warn = 1)

# Set ggplot theme
set_theme(theme_bw())
```

```
#' FUNCTION - aquaculture_eez_eval()
#'
#' This function generates a map and table showing the US West Coast Exclusive Economic
#' Zones (EEZ) colored and ordered, respectively, by the amount of area "suitable" for
#' marine aquaculture in each zone (km^2) for a single species. "Suitability" is defined
#' by Sea Surface Temperature (SST) (degC) and water depth (m).
```

```r
#'
#' @param species (Character) Species name
#' @param min_sst (Numeric) Minimum suitable SST for species (degC)
#' @param max_sst (Numeric) Maximum suitable SST for species (degC)
#' @param min_depth (Numeric) Minimum suitable depth for species (m). Should be <=0.
#' @param max_depth (Numeric) Maximum suitable depth for species (m). Should be <=0.
#' @return A results list with a tmap map object and priority summary table
#' @examples
#' aquaculture_eez_eval("Oysters", 11, 30, -70, 0)

aquaculture_eez_eval <- function(species, min_sst, max_sst, min_depth, max_depth) {

    # 1. IDENTIFY SUITABLE LOCATIONS

    # Set matrix for classify categories - tempC
    rcl_matrix_tempC <- matrix(c(
                        min_sst, max_sst, 1
                        ), ncol = 3, byrow = TRUE)

    # Reclassify tempC layer using tempC-defined suitability matrix
    oyster_tempC_zones <- classify(ocean_raster[[1]], rcl = rcl_matrix_tempC, others = 0)

    # Set matrix for classify categories - depthm
    rcl_matrix_depthm <- matrix(c(
                        min_depth, max_depth, 1
                        ), ncol = 3, byrow = TRUE)

    # Reclassify depthm layer using depthm-defined suitability matrix
    oyster_depthm_zones <- classify(ocean_raster[[2]], rcl = rcl_matrix_depthm, others = 0)

    # Multiply layers to classify by both tempC and depthm
    # Note: Only 1*1 cells will stay 1
    habitat_class <- oyster_tempC_zones * oyster_depthm_zones

    # Rename label
    names(habitat_class) <- "Suitable_Habitat"

    # 2. DETERMINE MOST SUITABLE EEZ

    # Turn eez vector data into raster
    eez_raster <- rasterize(eez, habitat_class, field = "rgn")
```

```
# Confirm rasterization of eez matches habitat_class raster
# Warning - confirm eez_raster and SST raster match before proceeding
# Note: compare crs, resolution, and extent
print("Checking that rasters match in CRS, resolution, and extent...")
if (same.crs(habitat_class,eez_raster)){
    # CRS match - true
    print("CRS do match!")
    if (all(res(habitat_class) == res(eez_raster))){
        # Resolution match - true
        print("Resolutions do match!")
        if (ext(habitat_class) == ext(eez_raster)) {
            # Extent match - true
            print("Extents do match!")
            print("Success - rasters match in all required characteristics.")
        } else {
            # Extent match - false
            stop("Extents do NOT match!")
        }
    } else {
        # Resolution match - false
        stop("Resolutions do NOT match!")
    }
} else {
    # CRS match - false
    stop("CRS do NOT match!")
}

# Grid cell area needs to be determined since crs is EPSG:4326 (units = deg, not km)
# Note: m^2 is default unit for cellSize()
habclass_cellarea <- cellSize(habitat_class, unit = "km")

# Multiply each cell in habitat_class by the cellSize areas
# Note: Because unsuitable = 0, unsuitable cells area will = 0
# Note: Only suitable cells (1) will keep the size value (km^2)
habitat_class_area <- habitat_class * habclass_cellarea

# Rename to add units
names(habitat_class_area) <- "Suitable_Habitat_km2"

# Calculate sum of suitable area in each EEZ using zonal operation
# habitat_class_area = raster where every suitable pixel represents size in km^2
# eez_raster = raster version of EEZ vector (EEZ regions)
```

```r
eez_priority <- zonal(habitat_class_area, eez_raster, fun = "sum", na.rm = TRUE)

# Sort eez regions by descending area
eez_priority <- eez_priority |>
    arrange(desc(Suitable_Habitat_km2))

# 3. GENERATE FIGURE

# Set color palette
color_palette <- c("#d9d9d9", "#0077b6")

# Add back EEZ polygon geometries for plotting
eez_mapv <- eez |>
    left_join(eez_priority, by = "rgn") |>
    # Sort by desc. suitable area
    arrange(desc(Suitable_Habitat_km2))

# Create EEZ region map colored by amount of suitable area
# Add basemap
map1 <- tm_basemap(server = "Esri.WorldTopoMap") +
# Add EEZ polygons
tm_shape(eez_mapv) +
tm_polygons(fill = "Suitable_Habitat_km2",
            fill_alpha = 0.6, # Transparency
            col = "#36454F", # Border color
            # Define color palette for 'aquaculture suitability' fill
            fill.scale = tm_scale_continuous(values = color_palette),
            # Re-label fill legend and adjust position
            fill.legend = tm_legend(title = "Suitable Area (km^2)",
                                    position = tm_pos_out("right","center"))) +
# Add region key as text inside each EEZ polygon
tm_text(text = "rgn_key",
        size = 0.8,
        col = "#36454F") +
# Add compass and scalebar
tm_compass(position = c("left","bottom")) +
tm_scalebar(position = c("left","bottom")) +
# Adjust pdf figure sizing - turn off autoscale
tm_options(component.autoscale = FALSE) +
# Add figure title
tm_title(paste("US West Coast EEZs:\nMost Suitable Area for",
               toTitleCase(species),"Aquaculture"),
```

```
                size = 1.8,
                frame = TRUE) +
    # Adjust legend text, fonts, and margins
    tm_layout(
            # Specify legend title size
            legend.title.size = 1.0,
            # Specify legend text size
            legend.text.size = 0.8,
            # Set font family
            text.fontfamily = "sans",
            # Set inner margins to give scalebar room
            inner.margins = c(0.1,0.1,0.05,0.05)
            )

    # 4. GENERATE TABLE

    # Make sorted table of prioritzed EEZs for this species
    priority_table <- eez_mapv |>
    select(rgn, rgn_key, Suitable_Habitat_km2) |>
    st_drop_geometry()

    # Return the map for this species
    return(list(map1,priority_table))
}
```

Prepare data

```
# LOAD DATA

# Load west coast EEZ vector file
eez <- sf::st_read(here("data","wc_regions_clean.shp"))
```

```
Reading layer `wc_regions_clean' from data source
  `/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/Aquaculture_Potential_
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

```r
# Load bathymetry raster
bath <- rast(here("data","depth.tif"))

# Load sea surface temp (sst) rasters
sst_08 <- rast(here("data","average_annual_sst_2008.tif"))
sst_09 <- rast(here("data","average_annual_sst_2009.tif"))
sst_10 <- rast(here("data","average_annual_sst_2010.tif"))
sst_11 <- rast(here("data","average_annual_sst_2011.tif"))
sst_12 <- rast(here("data","average_annual_sst_2012.tif"))

# Stack sst rasters
sst_stack <- c(sst_08,sst_09,sst_10,sst_11,sst_12)

# Check resolution and extent of sst_stack
cat("=== Sea Surface Temp Stack (K) ===\n")
```

```
=== Sea Surface Temp Stack (K) ===
```

```r
cat("Resolution:", res(sst_stack), "\n")
```

```
Resolution: 0.04166185 0.04165702
```

```r
cat("Extent:    "); print(ext(sst_stack))
```

```
Extent:
```

```
SpatExtent : -131.98475232999999, -114.98671702755556, 29.993052252666669, 49.98842296400000
```

```r
# Check resolution and extent of bath
cat("=== Bathymetry (m) ===\n")
```

```
=== Bathymetry (m) ===
```

```r
cat("Resolution:", res(bath), "\n")
```

```
Resolution: 0.004166667 0.004166667
```

```
cat("Extent:     "); print(ext(bath))
```

```
Extent:
```

```
SpatExtent : -132, -114, 28.999999999999996, 50 (xmin, xmax, ymin, ymax)
```

```
# Warning - CRS - confirm all three crs match, otherwise stop
if (same.crs(eez,bath) & same.crs(sst_stack, eez)){
    # All three CRS match - true
    print("SST Stack, West Coast EEZ, and Bathymetry data CRS all match!")
} else {
    # CRS match - false
    stop("CRS do NOT match!")
}
```

```
[1] "SST Stack, West Coast EEZ, and Bathymetry data CRS all match!"
```

```
# All CRS match, but EPSG code is missing from sst_stack WKT.
# Assign ESPG:4326 code to sst_stack WKT to avoid errors
crs(sst_stack) <- "EPSG:4326"
```

Process data

```
# Create raster with 1-layer of mean SST degC from 5-layer raster
sst_mean <- app(sst_stack, fun = "mean", na.rm = TRUE)
names(sst_mean) <- "Mean_SST"

# Check input and output raster layers to confirm
cat("Original SST stacked raster has:",nlyr(sst_stack),"layers","\n")
```

```
Original SST stacked raster has: 5 layers
```

```
cat("New SST Mean raster has:",nlyr(sst_mean),"layer","\n")
```

```
New SST Mean raster has: 1 layer
```

```r
# Check units of SST_mean raster
cat("Original range of Mean SST (K) raster:",minmax(sst_mean))
```

```
Original range of Mean SST (K) raster: 278.13 306.045
```

```r
# Convert SST_mean K to C (local operation - each cell)
sst_mean_C <- sst_mean - 273.15

# Confirm units of SST_mean_C raster
cat("Transformed range of Mean SST (degC) raster:",minmax(sst_mean_C))
```

```
Transformed range of Mean SST (degC) raster: 4.98 32.895
```

```r
# Crop the bath raster to the extent of the sst_mean_C raster
# Note: CRS must match
bath_crop <- crop(bath, sst_mean_C)
```

```r
# Resample bath raster to match sst_mean_C resolution using bilinear approach
bath_resampled <- resample(bath_crop, sst_mean_C, method = "bilinear")
```

```r
# Warning - confirm bathymetry and SST raster match before proceeding
# Note: compare crs, resolution, and extent
print("Checking that rasters match in CRS, resolution, and extent...")
```

```
[1] "Checking that rasters match in CRS, resolution, and extent..."
```

```r
if (same.crs(bath_resampled,sst_mean_C)){
    # CRS match - true
    print("CRS do match!")
    if (all(res(sst_mean_C) == res(bath_resampled))){
        # Resolution match - true
        print("Resolutions do match!")
        if (ext(sst_mean_C) == ext(bath_resampled)) {
            # Extent match - true
            print("Extents do match!")
            print("Success - rasters match in all required characteristics.")
        } else {
            # Extent match - false
            stop("Extents do NOT match!")
```

```
        }
    } else {
        # Resolution match - false
        stop("Resolutions do NOT match!")
    }
} else {
    # CRS match - false
    stop("CRS do NOT match!")
}
```

```
[1] "CRS do match!"
[1] "Resolutions do match!"
[1] "Extents do match!"
[1] "Success - rasters match in all required characteristics."
```

```
# Stack sst_mean_C and bath_resampled rasters
ocean_raster <- c(sst_mean_C,bath_resampled)
# Confirm stack
cat("New ocean_raster has",nlyr(ocean_raster),"layers.\n")
```

```
New ocean_raster has 2 layers.
```

```
cat("Layers are:",names(ocean_raster)[nlyr(ocean_raster)-1],",",names(ocean_raster)[nlyr(ocea
```

```
Layers are: Mean_SST , depth
```

Determine Most Suitable US West Coast EEZs for Marine Aquaculture

Species: Oysters

```
# Use aquaculture_eez_eval() to create map and table
oyster_results <- aquaculture_eez_eval("Oysters", 11, 30, -70, 0)
```

```
[1] "Checking that rasters match in CRS, resolution, and extent..."
[1] "CRS do match!"
[1] "Resolutions do match!"
[1] "Extents do match!"
[1] "Success - rasters match in all required characteristics."
```

```
# Display map
oyster_results[[1]]
```

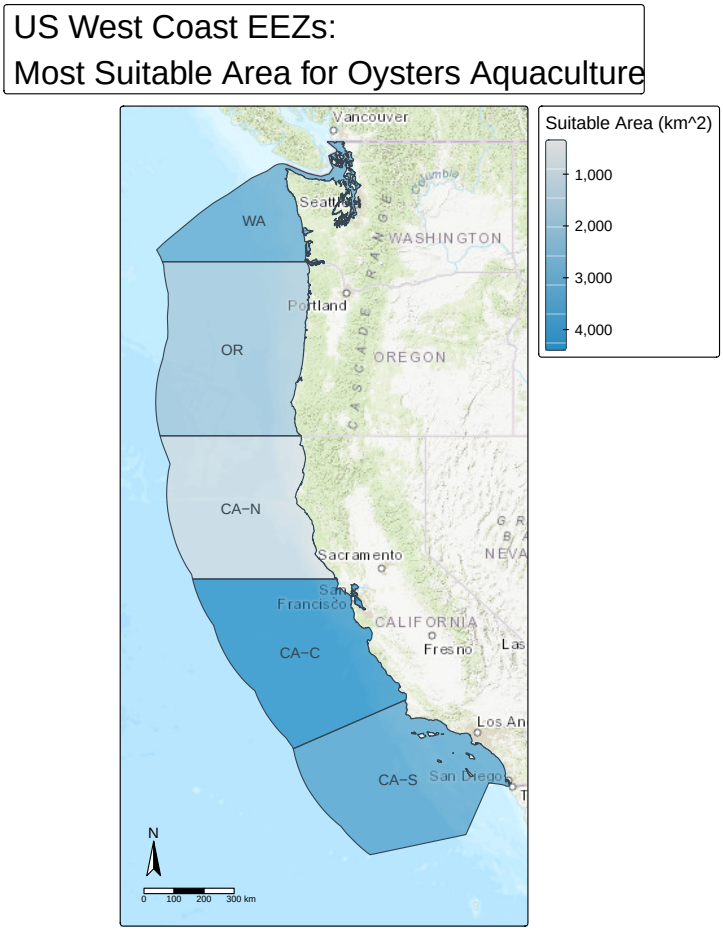


Figure 1: Map of the US West Coast Exclusive Economic Zones (EEZ), colored by amount of suitable area (km^2) for oysters marine aquaculture. Suitable area for oysters aquaculture is defined by areas meeting both the temperature requirement of 11 to 30 degrees C, and the depth requirement of 0 to 70 m.

```
    # Make table for this species using kable
    knitr::kable(oyster_results[[2]],
        digits = 2,
        col.names = c("EEZ Region", "Region Key", "Total Suitable Habitat (km^2)"))
```

Table 1: Total suitable area (km^2) for oysters marine aquaculture in each US West Coast Exclusive Economic Zone (EEZ). Suitable area for oysters marine aquaculture is defined by areas meeting both the temperature requirement of 11 to 30 degrees C, and the depth requirement of 0 to 70 m.

EEZ Region	Region Key	Total Suitable Habitat (km^2)
Central California	CA-C	4373.56
Southern California	CA-S	3329.90
Washington	WA	3091.52
Oregon	OR	1395.73
Northern California	CA-N	323.91

Species: Iceland scallop

```
# Use aquaculture_eez_eval() to create map and table
# Note: Iceland scallop temp range was ?-15 deg C, interpreted here as 0-15 degC
scallop_results <- aquaculture_eez_eval("Iceland Scallop", 0, 15, -250, -4)
```

```
[1] "Checking that rasters match in CRS, resolution, and extent..."
[1] "CRS do match!"
[1] "Resolutions do match!"
[1] "Extents do match!"
[1] "Success - rasters match in all required characteristics."
```

```
# Display map
scallop_results[[1]]
```

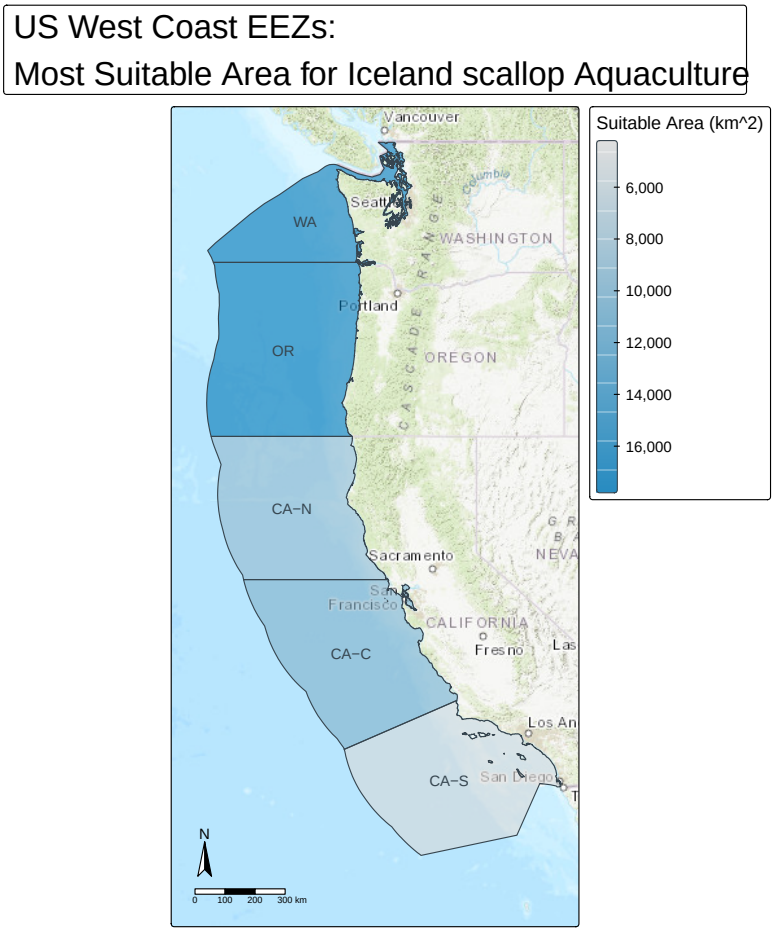


Figure 2: Map of the US West Coast Exclusive Economic Zones (EEZ), colored by amount of suitable area (km^2) for Iceland scallop (*Chlamys islandica*) marine aquaculture. Suitable area for Iceland scallop marine aquaculture is defined by areas meeting both the temperature requirement of 0 to 15 degree C, and the depth requirement of 4 to 250 m.

```
    # Make table for this species using kable
    knitr::kable(scallop_results[[2]],
        digits = 2,
        col.names = c("EEZ Region", "Region Key", "Total Suitable Habitat (km^2)"))
```

Table 2: Total suitable area (km^2) for Iceland scallop marine aquaculture in each US West Coast Exclusive Economic Zone (EEZ). Suitable area for Iceland scallop marine aquaculture is defined by areas meeting both the temperature requirement of 0 to 15 degrees C, and the depth requirement of 4 to 250 m.

EEZ Region	Region Key	Total Suitable Habitat (km^2)
Oregon	OR	17614.63
Washington	WA	17300.07
Central California	CA-C	9682.72
Northern California	CA-N	8420.54
Southern California	CA-S	4208.66

Reflection

In this analysis, the US West Coast Exclusive Economic Zones (EEZs) were assessed to determine the most suitable EEZs for expanding marine aquaculture. "Suitability" of an EEZ for marine aquaculture of a species was determined by calculating the total amount of area (km^2) in that EEZ that met the temperature and depth requirements for that species. Two potential marine aquaculture species were explored: oysters, with a temperature range of 11 to 30 degC and a depth range of 0 to 70 m, and Iceland scallops, with a temperature range of "?" to 15 degrees and a depth range of 4 to 250 m (Palomares, et al.). The "?" lower end of the scallop temperature range was assumed to be 0 for the purposes of this analysis.

The Central California (CA-C) EEZ contained the most suitable area for oysters marine aquaculture with 4373.56 km^2, while Northern California (CA-N) contained the least with 323.91 km^2 (Figure 1, Table 1). Oregon (OR) only had slightly more suitable area with 1395.73 km^2, while Southern California (CA-S) and Washington (WA) had 3329.90 and 3091.52 km^2, respectively. For Iceland scallops marine aquaculture, the OR and WA EEZs contained the most suitable area at 17614.63 and 17300.07 km^2, respectively (Figure 2, Table 2). CA-C had the next most suitable area at 9682.72 km^2, followed by CA-N at 8420.54 km^2 and CA-S at 4208.66 km^2. Overall, there is more potential for Iceland scallop marine aquaculture than oysters marine aquaculture in US West Coast EEZs. If only one EEZ were chosen for both species based on this analysis, the CA-C EEZ potentially has the most suitable marine aquaculture area for both species.

Data

The species, SST, Bathymetry, and EEZ data used in this analysis was provided by the instructor for this EDS223 class, downloaded in previous years from the links below.

Flanders Marine Institute (2026): MarineRegions.org. https://www.marineregions.org/eez.php

GEBCO Compilation Group. (2022). GEBCO_2022 Grid [Data set]. British Oceanographic Data Centre, National Oceanography Centre, NERC, UK. https://doi.org/10.5285/e0f0bb80-ab44-2739-e053-6c86abc0289c.

NOAA Coral Reef Watch. (2018). Daily global 5km satellite sea surface temperature anomaly (Version 3.1). U.S. Department of Commerce, National Oceanic and Atmospheric Administration. https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta.php.

Palomares, M. L. D., & Pauly, D. (Eds.). (n.d.). Country species summary: Chlamys islandica in United States. SeaLifeBase. Retrieved November 5, 2026, from https://www.sealifebase.ca/country/CountrySpeciesSummary.php?c_code=840&id=47720.

References

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